SCHEMA LINK: https://dbdiagram.io/d/6aad6b426d586d617e6bd187

1. Source Management

1a. data_source

Description: External provider/source identity. Stores where data comes from rather than the fetched data itself.

JSON


```
{
 "source_id": "8b7d8f2e-2d1a-4a91-9e10-7d2c5a1f0011",
 "source_code": "usgs",
 "name": "USGS Earthquake Hazards Program",
 "source_type": "api",
 "provider": "United States Geological Survey",
 "base_url": "https://earthquake.usgs.gov",
 "license": "USGS public domain",
"config": { "timeout_seconds": 30, "retry_count": 3, "rate_limit_per_minute": 60 },
 "is_active": true,
 "last_success_at": "2026-09-29T05:20:31Z",
 "last_failure_at": null,
 "created_at": "2026-09-29T05:00:00Z"
}
```


NOTE: One row identifies a provider/feed. Individual fetched responses belong to collection/raw_object.

1b. dataset

Description: Logical dataset or feed belonging to a source. A single source can expose multiple datasets.

JSON


```
{
  "dataset_id": "4a8e6f21-72d1-4f92-8c13-9b5500300011",
  "source_id": "8b7d8f2e-2d1a-4a91-9e10-7d2c5a1f0011",
  "dataset_code": "usgs_all_earthquakes",
  "name": "USGS All Earthquakes",
  "kind": "feed",  // static_reference, benchmark
  "modality": "tabular", // api, rss, file, social_media….
  "description": "USGS earthquake event feed",
  "is_active": true,
  "created_at": "2026-09-29T05:05:00Z"
}
```


NOTE: The dataset is the logical collection; versions/snapshots are represented by dataset_version.

1c. source_endpoint

Description: Represents one specific endpoint/feed exposed by a logical dataset. One dataset can expose multiple endpoints. Endpoint-specific configuration belongs here. Schema is optional and is represented through dataset_version when the endpoint has a defined schema/contract.


```
{
  "endpoint_id": "e81f3c92-6d47-4e29-a671-100000000011",
  "dataset_id": "4a8e6f21-72d1-4f92-8c13-9b5500300011",
  "endpoint_code": "earthquake_all_day",
  "name": "USGS All Day Earthquakes",
  "path": "/earthquakes/feed/v1.0/summary/all_day.geojson",
  "method": "GET",
  "response_format": "geojson",
  "config": {
    "query_params": {},
    "headers": {},
    "pagination": {
      "type": "none"
    },
    "response": {
      "format": "Application/json",
      },
    "polling": {
      "interval_seconds": 300
    }
  },
  "is_active": true,
  "created_at": "2026-09-29T05:05:00Z"
}
```


1d. dataset_version (optional, endpoint-specific schema/contract version)

Description: Optional schema/contract version for a specific endpoint. A dataset can have multiple endpoints, and each endpoint can have zero or more schema versions. A new version is created for an intentional schema/contract change, not merely because record counts changed.

JSON


```
{
  "dataset_version_id": "7f1a4c82-9e31-4b62-8d44-4c6600400011",
  "endpoint_id": "e81f3c92-6d47-4e29-a671-100000000011",

  "version": "1.0.0",

  "schema_version": "1.0.0",
  "schema_hash":
"sha256:8f6c2d91b7e45a3c9d1e7f204b6a8c51d3e9f2a4c7b1d6e8f0a3c5b7d9e2f411",

  "schema_definition": {
```

```
    "format": "geojson",
    "fields": {
      "id": {
        "type": "string",
        "required": true
      },
      "properties": {
        "type": "object",
        "required": true
      },
      "geometry": {
        "type": "object",
        "required": true
      }
    }
  },

  "released_at": "2026-09-29T05:00:00Z",
  "valid_from": "2026-09-29T05:00:00Z",
  "valid_to": null,

  "sample_count": 1248,

  "load_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",

  "created_at": "2026-09-29T05:21:00Z"
}
```

NOTE: dataset_version is optional. It is created only when an endpoint has a defined schema/contract that needs versioning. Record-count changes alone do not create a new version. The endpoint is the authoritative parent for the schema version.

2. Ingestion & Provenance

2a. Pipeline_run

Description: One execution of an ingestion or processing pipeline.

JSON

```
{
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "pipeline_name": "usgs_earthquake_ingestion",
  "pipeline_version": "1.0.0",
  "run_type": "scheduled",
  "status": "completed",
  "started_at": "2026-09-29T05:00:00Z",
  "completed_at": "2026-09-29T05:02:14Z",
  "records_read": 1284,
  "records_written": 1267,
```

```
  "records_rejected": 17,
  "error_message": null,
  "created_at": "2026-09-29T05:02:14Z"
}
```


NOTE: Tracks execution status, counts, configuration/version and failures for reproducibility and operations.

2b. Collection

Description: One actual request/fetch attempt against a configured source endpoint.

JSON


```
{
  "collection_id": "c9d52a71-83f4-4e29-a671-3d8200000011",
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "endpoint_id": "e81f3c92-6d47-4e29-a671-100000000011",
  "method": "GET",
  "request_uri": "https://earthquake.usgs.gov/earthquakes/feed/v1.0/summary/all_day.geojson",
  "http_status": 200,
  "started_at": "2026-09-29T05:00:11Z",
  "completed_at": "2026-09-29T05:00:13Z",
  "records_received": 1284,
  "status": "success",
  "error_message": null,
  "created_at": "2026-09-29T05:00:13Z"
}
```


NOTE: request_uri is where the data was fetched from; it is not the storage location of the fetched object. The endpoint determines the source and logical dataset for the collection.

2c. raw_object

Description: Immutable metadata for the exact raw payload stored in MinIO/S3-compatible object storage.

JSON


```
{
  "raw_object_id": "f21c8e73-4b91-45d6-a823-5e9400000011",
  "collection_id": "c9d52a71-83f4-4e29-a671-3d8200000011",
  "storage_uri": "s3://raw/usgs/2026/09/29/c9d52a71-response.json",
  "content_type": "application/geo+json",
  "size_bytes": 582941,
  "sha256": "8f6c2d91b7e45a3c9d1e7f204b6a8c51d3e9f2a4c7b1d6e8f0a3c5b7d9e2f411",
  "captured_at": "2026-09-29T05:00:13Z",
  "created_at": "2026-09-29T05:00:14Z"
}
```


NOTE: Raw bytes are immutable. Validation opinions belong in raw_validation so the same raw object can be revalidated with another validator version.

2d. raw_validation

Description: Validation result for a raw object under a particular endpoint schema/contract version and validator version. Schema validation is optional when the endpoint has no registered schema.

JSON


```
{
  "raw_validation_id": "a63f9271-52c8-4e14-9b73-6d8500000011",
  "raw_object_id": "f21c8e73-4b91-45d6-a823-5e9400000011",
  "dataset_version_id": "7f1a4c82-9e31-4b62-8d44-4c6600400011",
  "status": "passed",
  "schema_valid": true,
  "checksum_valid": true,
  "validator_version": "1.0.0",
  "validated_at": "2026-09-29T05:00:15Z"
}
```


NOTE: dataset_version_id is nullable. If an endpoint has a registered schema, validation can reference the exact schema version used. If the endpoint has no schema, schema_valid can be null while checksum/content validation can still be performed. A raw object can be revalidated later with another validator version without modifying the raw source truth.

2e. ingest_rejection

Description: Stores a source record that was rejected during ingestion/validation, together with the reason and details.

JSON


```
{
  "ingest_rejection_id": "d71b4f92-6e35-4a81-9c27-8f9600000011",
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "raw_object_id": "f21c8e73-4b91-45d6-a823-5e9400000011",
  "record_identifier": "usgs-20260929-001284",
  "stage": "validation",
  "rejection_code": "INVALID_COORDINATES",
  "reason": "Latitude outside valid range",
  "details": {
    "latitude": 194.27,
    "longitude": 84.12
  },
  "created_at": "2026-09-29T05:00:16Z"
}
```


NOTE: Rejected records remain traceable and replayable instead of silently disappearing.

3. Reference Data

3a. hazard

Description: Controlled vocabulary and hierarchy of hazard types.

JSON


```
{
  "hazard_id": "7a91c2e4-4d83-46b7-91f2-100000000011",
  "code": "earthquake",
  "name": "Earthquake",
  "category": "geophysical",
  "description": "Sudden ground shaking caused by the release of energy in the Earth's
crust.",
  "created_at": "2026-09-29T05:00:00Z"
}
```


Description: Controlled vocabulary and hierarchy of hazard types.

NOTE: Hazard is a reference concept, not a real-world disaster event. Hierarchies can represent parent/child hazards.

3b. admin_region

Description: Administrative geographic boundary reference data such as country, state, district, or sub-district.

JSON


```
{
  "admin_region_id": "2c71e8a5-6f34-4d92-8b15-200000000011",
  "dataset_version_id": "7f1a4c82-9e31-4b62-8d44-4c6600400011",
  "region_type": "district",
  "region_code": "BR-PAT",
  "name": "Patna",
  "parent_region_id": "5e82a1c7-3d64-4f91-9b27-300000000011",
  "geometry": "MULTIPOLYGON(...)",
  "centroid": "POINT(85.1376 25.5941)",
  "properties": {
    "state_code": "BR",
    "country_code": "IN"
  },
  "created_at": "2026-09-29T05:10:00Z"
}
```


NOTE: Stores reusable administrative boundaries. It is not an event-specific location table.

3c. static_raster

Description: Metadata and object-storage location for static geospatial reference rasters such as DEM, land cover, soil, and hazard maps.

JSON


```
{
  "static_raster_id": "9f42c731-8e15-4b62-a936-400000000011",
  "name": "SRTM Digital Elevation Model",
  "dataset_version_id": "7f1a4c82-9e31-4b62-8d44-4c6600400011",
  "raster_type": "dem",
  "storage_uri": "s3://reference/dem/srtm/india.tif",
```

```
  "crs": "EPSG:4326",
  "resolution": 30,
  "width": 12000,
  "height": 9000,
  "bounds": {
    "min_lon": 68.0,
    "min_lat": 6.0,
    "max_lon": 97.0,
    "max_lat": 37.0
  },
  "nodata_value": -9999,
  "created_at": "2026-09-29T05:20:00Z"
}
```


NOTE: Pixels remain in object storage; this table stores metadata needed to discover and use the raster.

3d. geo_feature

Description: Reusable geographic features such as rivers, roads, buildings, bridges, airports, or other GIS features.

JSON


```
{
  "geo_feature_id": "6d81f3a2-5c74-49e8-a271-500000000011",
  "feature_type": "river",
  "name": "Ganga River",
  "source_id": "8b7d8f2e-2d1a-4a91-9e10-7d2c5a1f0011",
  "external_id": "osm-way-123456789",
  "geometry": "LINESTRING(...)",
  "properties": {
    "waterway": "river"
  },
  "created_at": "2026-09-29T05:30:00Z"
}
```


NOTE: Provides reusable non-administrative GIS context for spatial analysis and impact assessment.

4. Canonical Source Data

4a. observation

Description: Normalized canonical representation of one source record. It is the main source-truth entity after source-specific parsing.

JSON


```
{
  "observation_id": "a81f3c92-6d47-4e15-9b21-100000000011",
  "dataset_version_id": "7f1a4c82-9e31-4b62-8d44-4c6600400011",
  "raw_object_id": "f21c8e73-4b91-45d6-a823-5e9400000011",
  "source_record_id": "usgs-20260929-001284",
```

```
  "hazard_id": "7a91c2e4-4d83-46b7-91f2-100000000011",
  "observed_at": "2026-09-29T04:58:21Z",
  "latitude": 25.5941,
  "longitude": 85.1376,
  "geometry": "POINT(85.1376 25.5941)",
  "value": {
    "magnitude": 4.8,
    "depth_km": 10.2,
    "place": "Patna region"
  },
  "quality_status": "valid",
  "created_at": "2026-09-29T05:01:00Z"
}
```


NOTE: Stores normalized source observations while retaining pointers to the raw object and dataset version.

4b. observation_region

Description: Associates an observation with one or more administrative regions using a spatial or semantic relationship.

JSON


```
{
  "observation_id": "a81f3c92-6d47-4e15-9b21-100000000011",
  "admin_region_id": "2c71e8a5-6f34-4d92-8b15-200000000011",
  "relation": "contains_point",
  "is_primary": true,
  "method": "point_in_polygon",
  "confidence": 0.99,
  "derived_result_id": null,
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "created_at": "2026-09-29T05:10:00Z"
}
```


NOTE: This avoids copying administrative region data into every observation.

4c. observation_relation

Description: Stores durable relationships between canonical observations, such as duplicate, superseding, correcting, retracting, reparsing, or corroborating relationships.

JSON


```
{
  "relation_id": "d91f3a72-6c84-4e25-8b17-600000000011",
  "observation_id": "a81f3c92-6d47-4e15-9b21-100000000011",
  "related_observation_id": "b92f4a83-7d95-4f36-8c28-700000000011",
  "relation_type": "duplicate_of",
  "method": "record_hash",
  "score": 0.99,
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "created_at": "2026-09-29T05:11:00Z"
}
```

NOTE: Temporary duplicate candidates should remain in memory; only the durable relationship is persisted.

4d. text_item

Description: Text-specific representation attached one-to-one to a text observation.

JSON


```
{
  "observation_id": "c81f3c92-6d47-4e15-9b21-800000000011",
  "text_kind": "news_article",
  "title": "Earthquake recorded near Patna",
  "body": "An earthquake was recorded in the Patna region...",
  "content_hash": "3e71c2a8f4b59d6e7c8a9b1d2e3f4051",
  "url": "https://example.org/news/earthquake",
  "author": "Reporter",
  "publisher": "Example News",
  "platform": "web",
  "parent_observation_id": null,
  "is_official": false,
  "engagement": {},
  "entities_raw": {}
}
```


NOTE: Keeps text-specific fields separate from the modality-neutral observation.

4e. raster_product

Description: Metadata for dynamic or observation-derived raster products such as satellite scenes, rainfall rasters, flood extent, or change maps.

JSON


```
{
  "observation_id": "d81f3c92-6d47-4e61-9b21-900000000011",
  "platform": "Sentinel-2",
  "sensor": "MSI",
  "product_id": "S2A_20260929_001",
  "processing_level": "L2A",
  "acquisition_start": "2026-09-29T04:30:00Z",
  "acquisition_end": "2026-09-29T04:40:00Z",
  "resolution_m": 10,
  "crs": "EPSG:4326",
  "cloud_cover_pct": 8.4,
  "polarization": null,
  "orbit_pass": "descending",
  "variables": {
    "ndvi": "available",
    "ndwi": "available"
  },
  "bands": [
    "B02",
    "B03",
    "B04",
    "B08"
  ],
```

```
  "is_pre_event": false,
  "storage_uri": "s3://derived/satellite/S2A_20260929_001.tif"
}
```

NOTE: Stores metadata and location of dynamic raster data. The large raster itself belongs in object storage.

4f. media

Description: Metadata for images, videos, and audio associated with observations.

JSON

```
{
  "media_id": "e81f3c92-6d47-4e61-9b21-a00000000011",
  "observation_id": "c81f3c92-6d47-4e61-9b21-800000000011",
  "media_type": "image",
  "source_url": "https://example.org/image.jpg",
  "raw_object_id": "f21c8e73-4b91-45d6-a823-5e9400000011",
  "raw_locator": "$.image_url",
  "perceptual_hash": "a83f71c92d",
  "width": 1920,
  "height": 1080,
  "duration_sec": null,
  "geometry": "POINT(85.1376 25.5941)",
  "captured_at": "2026-09-29T05:00:00Z"
}
```

NOTE: Supports image/video/audio metadata without duplicating binary media into PostgreSQL.

5. Enrichment & Derived Results

5a. model

Description: Identity and version information for an ML/AI model used to generate derived results.

JSON

```
{
  "model_id": "2f71c8a4-6d93-4e25-9b81-100000000011",
  "name": "DisasterMultimodalClassifier",
  "version": "1.0.0",
  "task": "disaster_classification",
  "framework": "pytorch",
  "architecture": "multimodal_transformer",
  "model_uri": "s3://models/disaster-classifier/1.0.0/model.pt",
  "training_dataset_version_id": "7f1a4c82-9e31-4b62-8d44-4c6600400011",
  "config": {
    "input_modalities": [
      "text",
      "image"
    ]
  },
  "is_active": true,
  "created_at": "2026-09-29T06:00:00Z"
}
```

NOTE: A model is versioned independently from its predictions.

5b. derived_result

Description: Durable output produced from ML inference, deterministic rules, spatial analysis, or human review.

JSON


```
{
  "derived_result_id": "5c82a1e7-4d63-49f2-8b31-200000000011",
  "observation_id": "a81f3c92-6d47-4e15-9b21-100000000011",
  "model_id": "2f71c8a4-6d93-4e25-9b81-100000000011",
  "result_type": "disaster_classification",
  "status": "completed",
  "result": {
    "label": "earthquake",
    "confidence": 0.94,
    "classes": {
      "earthquake": 0.94,
      "flood": 0.02,
      "landslide": 0.01,
      "other": 0.03
    }
  },
  "confidence": 0.94,
  "output_uri": "s3://derived/results/5c82a1e7.json",
  "supersedes_result_id": null,
  "created_at": "2026-09-29T06:05:00Z"
}
```


NOTE: Only durable derived outputs are persisted. Temporary tensors, candidate sets, similarity matrices, and intermediate computations stay outside the database.

5c. derived_result_input

Description: Lineage table showing which durable source objects, observations, events, or other results contributed to a derived result.

JSON


```
{
  "derived_result_input_id": "91e4c732-5a86-4d19-8b42-300000000011",
  "derived_result_id": "5c82a1e7-4d63-49f2-8b31-200000000011",
  "observation_id": "a81f3c92-6d47-4e15-9b21-100000000011",
  "media_id": null,
  "static_raster_id": null,
  "geo_feature_id": null,
  "input_result_id": null,
  "modality_event_id": null,
  "event_id": null,
  "input_role": "primary_observation",
  "created_at": "2026-09-29T06:05:00Z"
}
```

NOTE: Exactly one durable input target should be associated with each row.

6. Source Scoring

6a. source_score

Description: Contextual score for a source by hazard/task and score type, such as relevance, reliability, authority, or freshness.

JSON


```
{
  "source_score_id": "6f81c2a4-5d93-4e27-9b41-100000000011",
  "source_id": "8b7d8f2e-2d1a-4a91-9e10-7d2c5a1f0011",
  "hazard_id": "7a91c2e4-4d83-46b7-91f2-100000000011",
  "task": "earthquake_detection",
  "score_type": "reliability",
  "score": 0.94,
  "methodology": "historical_source_evaluation",
  "methodology_version": "1.0",
  "evidence": {
    "sample_count": 12480,
    "agreement_rate": 0.96
  },
  "valid_from": "2026-09-29T00:00:00Z",
  "valid_to": null,
  "created_at": "2026-09-29T06:30:00Z"
}
```


NOTE: Scores are contextual rather than universal; a source can have different usefulness for different hazards and tasks.

7. Durable Event Intelligence

7a. modality_event

Description: Independently detected event within a particular evidence domain/modality.

JSON


```
{
  "modality_event_id": "me91e4c7-5a86-4d19-8b42-700000000011",
  "origin": "source_reported",
  "evidence_domain": "earthquake",
  "hazard_id": "7a91c2e4-4d83-46b7-91f2-100000000011",
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "start_time": "2026-09-29T04:58:18Z",
  "end_time": null,
  "time_precision": "second",
  "published_at": "2026-09-29T05:00:00Z",
  "geometry": "POINT(85.1376 25.5941)",
  "location_uncertainty_km": 2.0,
```

```
  "magnitude": 4.8,
  "magnitude_type": "Mw",
  "depth_km": 10.2,
  "severity": "moderate",
  "confidence": 0.95,
  "status": "active",
  "supersedes_modality_event_id": null,
  "attributes": {
    "source_record_id": "usgs-20260929-001284"
  },
  "created_at": "2026-09-29T06:40:00Z"
}
```

NOTE: A modality event is not yet the final unified disaster event.

7b. modality_event_input

Description: Connects a modality event to the observations or derived results that produced it.

JSON

```
{
  "modality_event_input_id": "mi91e4c7-5a86-4d19-8b42-800000000011",
  "modality_event_id": "me91e4c7-5a86-4d19-8b42-700000000011",
  "observation_id": "a81f3c92-6d47-4e15-9b21-100000000011",
  "derived_result_id": null,
  "input_role": "primary_observation",  // what role this input plays
  "weight": 1.0,
  "created_at": "2026-09-29T06:41:00Z"
}
```

NOTE: Provides durable lineage from modality event back to source evidence.

7c. event_match

Description: Stores the durable decision about whether two modality events represent the same real-world event.

JSON

```
{
  "event_match_id": "em91e4c7-5a86-4d19-8b42-900000000011",
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "modality_event_a_id": "me91e4c7-5a86-4d19-8b42-700000000011",
  "modality_event_b_id": "me72e8a5-6f34-4d92-8b15-900000000011",
  "overall_score": 0.93,
  "decision": same_event",
  "method": "machine_learning",
  "signals": {
    "temporal_similarity": 0.98,
    "spatial_similarity": 0.91,
    "semantic_similarity": 0.9
  },
  "created_at": "2026-09-29T06:45:00Z"
}
```

NOTE: Temporary candidate pairs should not be persisted; this table stores the durable matching decision.

7d. event

Description: Unified real-world disaster event formed by matching and fusing evidence from multiple modality events.

JSON


```
{
  "event_id": "5e16a8c2-9b47-4d65-8f54-500000000011",
  "event_code": "EQ-2026-09-29-001",
  "hazard_id": "7a91c2e4-4d83-46b7-91f2-100000000011",
  "name": "Patna Earthquake Event",
  "start_time": "2026-09-29T04:58:18Z",
  "end_time": null,
  "time_precision": "second",
  "centroid": "POINT(85.1376 25.5941)",
  "affected_area": null,
  "magnitude": 4.8,
  "magnitude_type": "Mw",
  "severity": "moderate",
  "confidence": 0.95,
  "confidence_method": "source_fusion",
  "status": "detected",
  "verification": "verified",
  "verification_notes": "Confirmed by multiple modality events.",
  "description": "Unified earthquake event assembled from multiple source
observations.",
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "created_at": "2026-09-29T07:00:00Z",
  "updated_at": "2026-09-29T07:00:00Z"
}
```


NOTE: This is the canonical unified disaster event. It is different from modality_event, which represents an individual detection.

7e. event_attribute_selection

Description: Records why a particular modality event/value was selected as the canonical event attribute when sources disagree.

JSON


```
{
  "event_attribute_selection_id": "ea91e4c7-5a86-4d19-8b42-a00000000011",
  "event_id": "5e16a8c2-9b47-4d65-8f54-500000000011",
  "attribute_name": "magnitude",
  "modality_event_id": "me91e4c7-5a86-4d19-8b42-700000000011",
  "value": {
    "magnitude": 4.8
  },
  "method": "highest_authority",
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "created_at": "2026-09-29T07:01:00Z"
}
```

NOTE: Preserves the decision logic behind canonical event attributes.

7f. event_member

Description: Connects modality events to their unified event.

JSON


```
{
  "event_id": "5e16a8c2-9b47-4d65-8f54-500000000011",
  "modality_event_id": "me91e4c7-5a86-4d19-8b42-700000000011",
  "membership_score": 0.96,
  "is_primary": true,
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "created_at": "2026-09-29T07:02:00Z"
}
```


NOTE: Allows one unified event to contain multiple modality-level detections.

7g. event_evidence

Description: Connects a unified event to supporting, contradicting, or contextual evidence.

JSON


```
{
  "event_evidence_id": "ee91e4c7-5a86-4d19-8b42-b00000000011",
  "event_id": "5e16a8c2-9b47-4d65-8f54-500000000011",
  "observation_id": "a81f3c92-6d47-4e15-9b21-100000000011",
  "derived_result_id": "5c82a1e7-4d63-49f2-8b31-200000000011",
  "stance": "supports",
  "confidence": 0.94,
  "is_primary": true,
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "created_at": "2026-09-29T07:03:00Z"
}
```


NOTE: Evidence is kept separate from event membership because evidence can support or contradict an event.

7h. event_region

Description: Associates a unified event with administrative regions and identifies primary affected regions.

JSON


```
{
  "event_id": "5e16a8c2-9b47-4d65-8f54-500000000011",
  "admin_region_id": "2c71e8a5-6f34-4d92-8b15-200000000011",
  "relation": "intersects",
  "overlap_fraction": 0.62,
  "is_primary": true,
  "method": "spatial_intersection",
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "created_at": "2026-09-29T07:04:00Z"
}
```

NOTE: Avoids duplicating region information inside event records.

7i. event_impact

Description: Stores durable impacts associated with an event, such as casualties, damage, displacement, affected population, area, or needs.

JSON


```
{
  "event_impact_id": "ei91e4c7-5a86-4d19-8b42-c00000000011",
  "event_id": "5e16a8c2-9b47-4d65-8f54-500000000011",
  "kind": "affected_area",
  "category": "estimated_affected_area",
  "origin": "estimated",
  "metric_value": 42.7,
  "unit": "km2",
  "admin_region_id": "2c71e8a5-6f34-4d92-8b15-200000000011",
  "geo_feature_id": null,
  "geometry": "POLYGON(...)",
  "confidence": 0.91,
  "is_canonical": true,
  "method": "satellite_change_detection",
  "source_observation_id": null,
  "source_derived_result_id": "5c82a1e7-4d63-49f2-8b31-200000000011",
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "created_at": "2026-09-29T07:10:00Z"
}
```


NOTE: This is where durable affected-area, damage, casualty, displacement, and needs information belongs.

7j. event_lineage

Description: Records how the platform's event representation changed, such as superseding, merging, or splitting.

JSON


```
{
  "event_lineage_id": "el91e4c7-5a86-4d19-8b42-d00000000011",
  "from_event_id": "5e16a8c2-9b47-4d65-8f54-500000000011",
  "to_event_id": "6f27b9d3-0c58-4e76-9a65-600000000011",
  "relation": "superseded_by",
  "reason": "New evidence changed the unified event boundary.",
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "created_at": "2026-09-29T07:20:00Z"
}
```


NOTE: Event lineage describes database/event-model evolution, not real-world disaster causality.

7k. event_relationship

Description: Represents a real-world relationship between two unified disaster events, such as triggered, caused, contributed_to, associated_with, or concurrent_with.

JSON


```
{
  "event_relationship_id": "er91e4c7-5a86-4d19-8b42-e00000000011",
  "parent_event_id": "EV-RAIN-001",
  "child_event_id": "EV-FLOOD-001",
  "relationship_type": "triggered",
  "confidence": 0.89,
  "evidence": {
    "time_gap_minutes": 14,
    "spatial_distance_km": 8.2
  },
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "created_at": "2026-09-29T07:30:00Z"
}
```


NOTE: This is distinct from event_lineage. It represents a relationship between real-world events, such as rainfall triggering flood or flood contributing to landslide.

7k. event_group

Description: Represents a broader disaster episode containing multiple distinct but related real-world events.

JSON


```
{
  "event_group_id": "ep-000205",
  "event_group_code": "EP-2026-MONSOON-000205",
  "name": "Example Monsoon Disaster Episode",
  "description": "A cascading disaster episode involving heavy rainfall, flooding,
landslides and impacts.",
  "start_time": "2026-08-10T00:00:00Z",
  "end_time": "2026-08-13T14:00:00Z",
  "centroid": "POINT(85.1376 25.5941)",
  "affected_area": "MULTIPOLYGON(...)",
  "severity": "high",
  "confidence": 0.89,
  "status": "active",
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "created_at": "2026-08-13T15:00:00Z",
  "updated_at": "2026-08-13T15:00:00Z"
}
```


NOTE: Groups distinct events into a broader disaster episode without collapsing them into one event.

7l. event_group_member

Description: Connects individual unified events to a disaster episode and records their role within that episode.

JSON


```
{
  "event_group_id": "ep-000205",
  "event_id": "EV-FLOOD-0001",
```

```
  "role": "primary",
  "is_primary": true,
  "confidence": 0.93,
  "pipeline_run_id": "b4f8c2a1-7d93-4e61-9b25-1a6e50000011",
  "created_at": "2026-08-13T15:05:00Z"
}
```

NOTE: Preserves each event's role within an event group, such as primary, trigger, related, consequence, or impact.

8. Benchmark / Evaluation

dataset_sample

Description: Represents one benchmark/training/evaluation sample and its event composition.

JSON

```
{
  "dataset_sample_id": "sample-000002",
  "dataset_version_id": "7f1a4c82-9e31-4b62-8d44-4c6600400011",
  "external_sample_id": "SAMPLE-000002",
  "sample_type": "multi_event",
  "event_group_id": "ep-000205",
  "split": "test",
  "label": {
    "primary_hazard": "flood",
    "related_hazards": [
      "landslide"
    ],
    "severity": "high"
  },
  "weight": 1.0,
  "created_at": "2026-09-28T10:00:00Z"
}
```

NOTE: The sample is the benchmark prediction/evaluation unit and can represent a single event or multiple related events.

dataset_sample_item

Description: Connects one sample to the actual canonical observation, media item, or derived result used as an input.

JSON

```
{
  "dataset_sample_item_id": "sample-item-0001",
  "dataset_sample_id": "sample-000002",
  "role": "text_input",
  "observation_id": "a81f3c92-6d47-4e15-9b21-100000000011",
  "media_id": null,
  "derived_result_id": null,
  "created_at": "2026-09-28T10:00:01Z"
}
```

NOTE: One sample can contain many items. The benchmark layer assembles existing canonical data instead of duplicating it.

dataset_sample_event

Description: Connects a benchmark sample to one or more unified events and identifies each event's role in the sample.

JSON


```
{
  "dataset_sample_event_id": "sample-event-0001",
  "dataset_sample_id": "sample-000002",
  "event_id": "EV-FLOOD-0001",
  "role": "primary",
  "is_primary": true,
  "relationship_to_primary": null,
  "confidence": 0.93,
  "created_at": "2026-09-28T10:00:02Z"
}
```


NOTE: Allows a benchmark sample to contain one or multiple events, including primary, trigger, related, consequence, and impact roles.

9. Final Expected Dataset

Description: Final dataset produced by the platform after source collection, raw storage, validation, normalization, deduplication, enrichment, event detection, event matching, event fusion, impact estimation, and benchmark preparation.

JSON


```
{
  "dataset": {
    "dataset_code": "multimodal_disaster_final_dataset",
    "version": "1.0",
    "created_at": "2026-09-29T10:00:00Z"
  },
  "sample": {
    "sample_id": "SAMPLE-000002",
    "sample_type": "multi_event",
    "event_group_id": "EP-2026-MONSOON-000205",
    "events": [
      {
        "event_id": "EV-RAIN-0001",
        "hazard": "heavy_rainfall",
        "role": "trigger",
        "start_time": "2026-08-10T02:00:00Z",
        "end_time": "2026-08-10T18:00:00Z",
        "location": {
          "latitude": 25.61,
          "longitude": 85.14
        }
      },
```

```
    {
      "event_id": "EV-FLOOD-0001",
      "hazard": "flood",
      "role": "primary",
      "start_time": "2026-08-10T19:00:00Z",
      "end_time": "2026-08-12T18:00:00Z",
      "location": {
        "latitude": 25.59,
        "longitude": 85.13
      }
    },
    {
      "event_id": "EV-LANDSLIDE-0001",
      "hazard": "landslide",
      "role": "related",
      "start_time": "2026-08-11T04:30:00Z",
      "end_time": "2026-08-11T09:00:00Z",
      "location": {
        "latitude": 25.64,
        "longitude": 85.2
      }
    }
  ],
  "event_relationships": [
    {
      "from_event": "EV-RAIN-0001",
      "to_event": "EV-FLOOD-0001",
      "relationship": "triggered",
      "confidence": 0.91
    },
    {
      "from_event": "EV-FLOOD-0001",
      "to_event": "EV-LANDSLIDE-0001",
      "relationship": "contributed_to",
      "confidence": 0.84
    }
  ],
  "inputs": {
    "earthquake": [],
    "weather": [
      {
        "observation_id": "OBS-WEATHER-001",
        "rainfall_mm": 186.4,
        "observed_at": "2026-08-10T18:00:00Z"
      }
    ],
    "text": [
      {
        "observation_id": "OBS-NEWS-001",
        "text_item_id": "TEXT-001",
        "title": "Heavy rainfall causes flooding in the district"
      }
    ],
    "images": [
      {
        "observation_id": "OBS-IMAGE-001",
        "media_id": "MEDIA-001",
        "media_type": "image"
```

```
        }
      ],
      "satellite": [
        {
          "observation_id": "OBS-SAT-001",
          "product_id": "S2A_20260811_001",
          "derived_result_id": "RESULT-FLOOD-001"
        }
      ]
    },
    "impact": {
      "affected_area_km2": 42.7,
      "affected_population": 185000,
      "casualties": 12,
      "displaced_population": 4300
    },
    "target": {
      "primary_hazard": "flood",
      "related_hazards": [
        "landslide"
      ],
      "severity": "high",
      "event_detected": true
    }
  }
}
```


NOTE: This is the expected final dataset shape. The actual dataset will contain many samples and may contain single-event or multi-event samples. Raw payloads and large media/raster objects remain in object storage; the final dataset references canonical observations, media, derived results, and unified events rather than duplicating those objects.

Expected output: Each final sample should be usable for disaster-event detection, hazard classification, severity prediction, event matching, multi-event prediction, cascading-event analysis, impact estimation, and multimodal fusion.

#	Table	New row on every pipeline run?	When is it created?
1	`data_source`	No	Once when registering a source/provider. Example: USGS
2	`dataset`	No	Once when registering a logical feed/dataset
3	`dataset_version`	Optional	Only when creating an intentional snapshot/release/version
4	`pipeline_run`	Yes	Every execution of an ingestion/processing pipeline
5	`collection`	Yes*	Every actual fetch/request attempt
6	`raw_object`	Usually	Every newly stored raw payload
7	`raw_validation`	Per validation	Whenever a raw object is validated/revalidated
8	`ingest_rejection`	Only on rejection	When a record fails ingestion/validation
9	`hazard`	No	Reference data; create once per hazard concept
10	`admin_region`	Usually	When loading/changing administrative boundary data
11	`static_raster`	No	When a new static raster/version is registered
12	`geo_feature`	Usually	When geographic reference features are loaded/updated
13	`observation`	Only for new/changed source records	When a source record is normalized into canonical form
14	`observation_region`	Per association	When an observation is associated with an administrative region

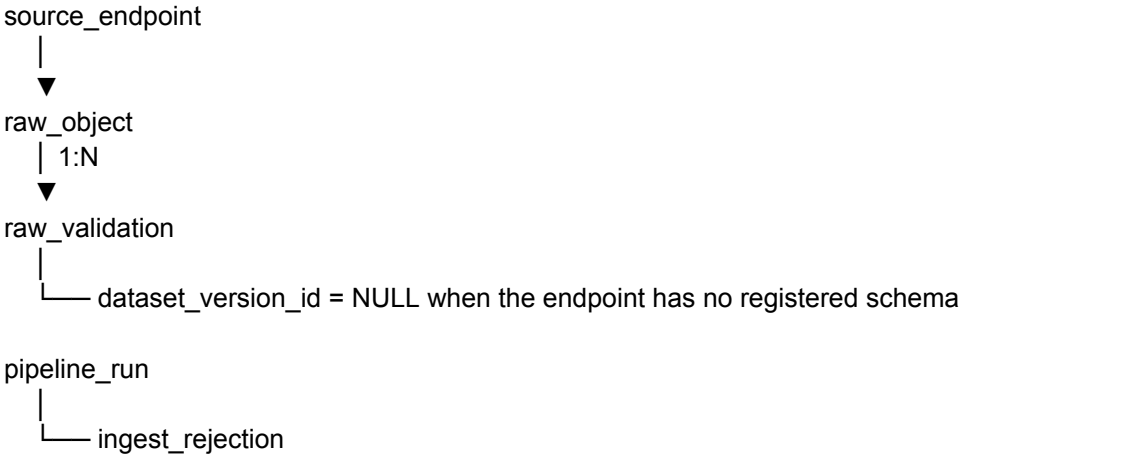
15	observation_relation	Only when relationship exists	Duplicate, correction, superseding, corroboration, etc.
16	text_item	Per text observation	When an observation contains text
17	raster_product	Per raster observation	When a raster product is associated with an observation
18	media	Per media item	When an observation contains image/video/audio
19	model	No	When a new model/version is registered
20	derived_result	Per inference/result	When ML/rules/spatial analysis produces a durable result
21	derived_result_input	Per input lineage	When a source contributes to a derived result
22	source_score	Usually	When source reliability/relevance/authority is evaluated or updated
23	modality_event	Per detected event	When an event is detected within a modality
24	modality_event_input	Per event input	When evidence is linked to a modality event
25	event_match	Per durable match decision	When two modality events are determined to be same/different
26	event	Per unified event	When modality evidence is fused into a real-world event
27	event_attribute_selection	Per selected attribute	When choosing the canonical value from competing evidence
28	event_member	Per membership	When a modality event belongs to a unified event
29	event_evidence	Per evidence relationship	When evidence supports/contradicts/contextualizes an event
30	event_region	Per region association	When an event is associated with an administrative region

31	`event_impact`	Per durable impact	When damage/casualty/displacement/affected-area information is produced
32	`event_lineage`	Only when event changes	Merge, split, supersede, etc.
33	`event_relationship`	Only when relationship exists	Rainfall → triggered → flood
34	`event_group`	Per disaster episode	When multiple related events form a broader episode
35	`event_group_member`	Per membership	When an event is assigned to an event group
36	`dataset_sample`	Per benchmark sample	When creating a training/validation/test sample
37	`dataset_sample_item`	Per sample input	When observations/media/results are included in a sample
38	`dataset_sample_event`	Per sample-event relationship	When an event is included in a benchmark sample

FINAL SOURCE / INGESTION RELATIONSHIP

data_source
  │ 1:N
  ▼
dataset
  │ 1:N
  ▼
source_endpoint
  │ 0:N
  ▼
dataset_version
  │
  │ optional endpoint schema/contract

pipeline_run
  │ 1:N
  ▼
collection
  │ N:1
  ▼



Core rule:
One dataset can have multiple endpoints. Each endpoint may have zero or more schema/contract
versions. Dataset-versioning is endpoint-specific and optional; changes in record count alone do not create a new schema version. Raw objects remain immutable, while validation results are stored separately and may be repeated with different validator versions.